Who Decides? From One Decision to a Spectrum of Tasks (Uncorrected Proofs)

ECON 470 — Microeconomics of Artificial Intelligence

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Spring 2026

1 Setup

A single decision must be made. There are N possible states of the world, $\theta \in \{1, 2, \ldots, N\}$. The prior probability of the default state $\theta = 1$ is $p > 1/N$; the remaining states share the residual mass equally. There are N possible actions; the agent earns R if the action matches the realized state and 0 otherwise.

There are three ways the decision can get made:

- **Rule** ("dumb machine"): always pick $a = 1$, the default action. No information is used. Per-decision payoff: pR.
- **Smart machine**: takes the AI's signal, picks the matching action. The AI predicts the state correctly with probability $e > p$. Coding the machine to map signals to actions costs C per decision. Per-decision payoff: $eR - C$.
- **Human (with AI)**: takes the AI's signal, picks the matching action — but with a one-period delay (discount $\delta < 1$) and an ability to catch and correct a fraction $\alpha \in [0, 1]$ of the AI's errors. Per-decision payoff: $\delta[e + (1 - e)\alpha]R$.

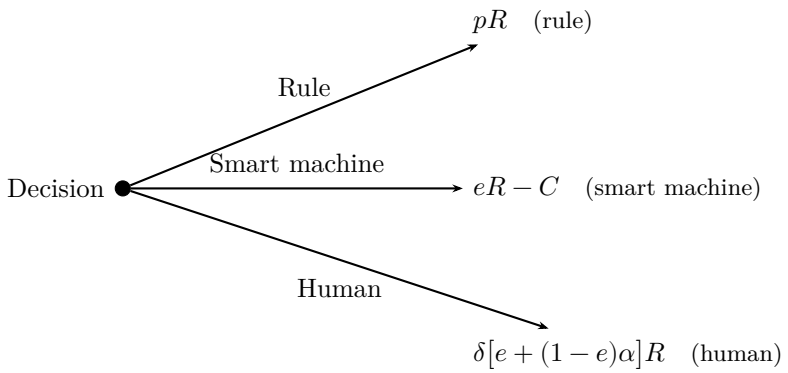
The human's payoff has two effects relative to the smart machine: a *speed disadvantage* ($\delta < 1$) and an *error-correction advantage* ($\alpha > 0$ means the human catches some of the AI's $(1 - e)$ errors). Whether the human or the machine is preferred depends on which effect dominates.

Parameters for all examples: $R = 100$, $p = 0.6$, $\delta = 0.75$, $\alpha = 0.7$, $C = 10$ (so $C/R = 0.10$). These are chosen so that all three regimes win in a wide range of e (Part I) and so that the human region in Part II has a clear biphasic pattern as the AI improves.

Part I: Gans's §6.3 — One Decision, Three Regimes

In this part we analyze a single decision at a single AI accuracy e. The question is: which of the three regimes is optimal? The answer depends on e, α, C/R, and δ. We will derive three boundary conditions, draw the "who decides" diagram, and trace what happens as the AI improves.

2 Three Regimes, One Decision



The agent picks whichever regime delivers the highest expected per-decision payoff. We now find the parameter regions in which each regime wins.

3 Boundary 1: Smart Machine vs. Rule

The smart machine beats the rule iff $eR - C \geq pR$, i.e.,

$$\boxed{\frac{C}{R} \leq e - p.}$$

The coding cost (relative to the prize) must not exceed the AI's accuracy improvement over the prior. If coding is too expensive, fall back to the rule. If the AI is too inaccurate (relative to the prior), fall back to the rule.

4 Boundary 2: Human vs. Rule

The human beats the rule iff $\delta R[e + (1-e)\alpha] \geq pR$, i.e.,

$$\boxed{\delta[e + (1-e)\alpha] \geq p.}$$

The human's accuracy with mitigation is $e + (1-e)\alpha$ (correct when AI is correct, plus a fraction α of the cases where AI is wrong). The human beats the rule when this discounted accuracy exceeds

the prior. Solving for the threshold e: the human beats the rule iff $e \geq (p/\delta - \alpha)/(1-\alpha)$. At our parameters this gives $e \geq 0.333$.

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5 Boundary 3: Smart Machine vs. Human

The smart machine beats the human iff $eR - C \geq \delta R[e + (1-e)\alpha]$. Rearranging:

$$\boxed{\frac{C}{R} \leq e(1-\delta) - \alpha\delta(1-e).}$$

This is the load-bearing boundary. The right-hand side is the smart machine's *speed premium* over the human ($e(1-\delta)$, the value of avoiding the delay) minus the human's *error-mitigation premium* over the smart machine ($\alpha\delta(1-e)$, the value of catching AI errors). The smart machine wins when the speed premium is bigger than the mitigation premium plus the coding cost.

Note the comparative static in e. The right-hand side is increasing in e:

$$\frac{\partial}{\partial e}[e(1-\delta) - \alpha\delta(1-e)] = (1-\delta) + \alpha\delta > 0.$$

As the AI gets better, the threshold for the smart machine to beat the human rises. **More territory for smart machines, less for humans.** This is Gans's headline §6.3 result.

The intuition is direct: the human's only advantage is mitigation, $\alpha\delta(1-e)R$. As $e \to 1$, there are fewer errors to catch, so the human's advantage shrinks to zero. In the limit, only the speed/coding-cost trade-off remains, and the smart machine dominates whenever coding is cheap.

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6 A Concrete Calculation

At our parameters ($p = 0.6$, $\delta = 0.75$, $\alpha = 0.7$, $C/R = 0.10$), we sweep e and compute all three regime payoffs.

e	Rule (pR)	Smart ($eR - C$)	Human ($\delta(e + (1-e)\alpha)R$)	Winner
0.20	60.00	10.00	57.00	**Rule**
0.30	60.00	20.00	59.25	Rule
0.40	60.00	30.00	61.50	Human
0.50	60.00	40.00	63.75	Human
0.60	60.00	50.00	66.00	Human
0.70	60.00	60.00	68.25	Human
0.80	60.00	70.00	70.50	Human

e	Rule (pR)	Smart ($eR - C$)	Human ($\delta(e + (1-e)\alpha)R$)	Winner
0.85	60.00	75.00	71.62	**Smart**
0.95	60.00	85.00	73.88	Smart

All three regimes appear. **For low e ($e < 0.33$), the rule wins** — the AI is too inaccurate to be worth using at all. **For moderate e ($0.33 \leq e \leq 0.81$), the human wins** — their mitigation skill more than compensates for the delay. **For high e ($e > 0.81$), the smart machine wins** — the human's mitigation advantage is no longer worth the delay.

The crossovers are exactly where the boundaries we derived put them:

- Rule beats human iff $e < 0.333$ (boundary 2).
- Smart machine beats human iff $e > 0.806$ (boundary 3).

This is the §6.3 displacement story in numbers. The mechanism: the human's advantage over the smart machine is the mitigation term $\alpha\delta(1-e) = 0.7 \cdot 0.75 \cdot (1-e) = 0.525(1-e)$, which is 0.42 at $e = 0.2$ but shrinks to 0.026 at $e = 0.95$. Once mitigation drops below the speed-and-coding-cost gap, the smart machine wins.

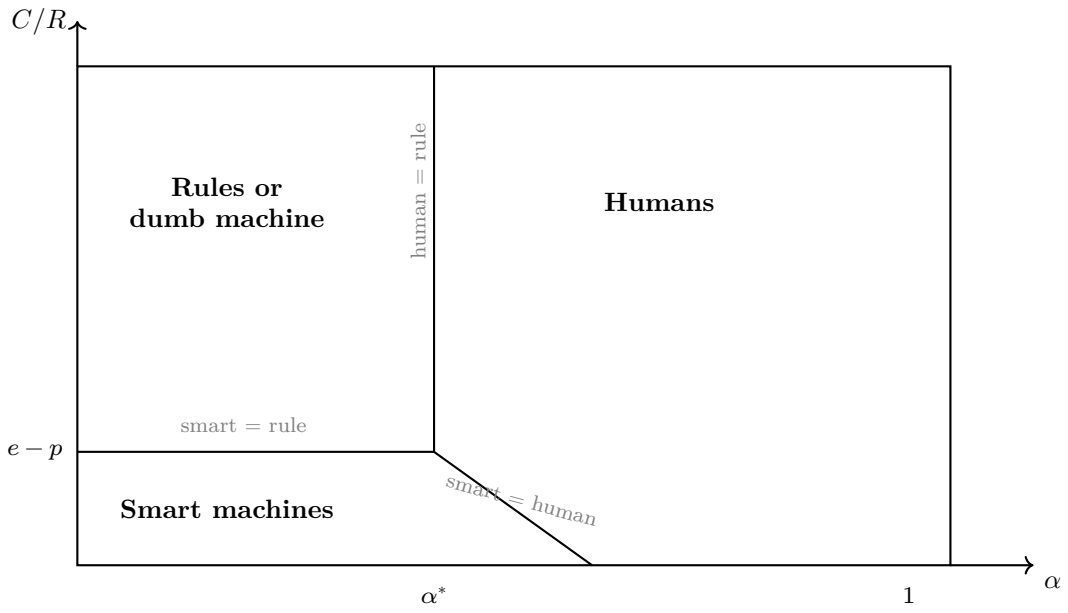
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7 Figure: Who Decides in $(\alpha, C/R)$ Space

The three boundaries can be plotted simultaneously in $(\alpha, C/R)$ space at fixed e, p, δ. Boundary 1 (smart vs. rule) is a horizontal line at $C/R = e - p$. Boundary 2 (human vs. rule) is a vertical line at $\alpha = (p - \delta e)/[\delta(1-e)]$. Boundary 3 (smart vs. human) is a downward-sloping line, $C/R = e(1-\delta) - \alpha\delta(1-e)$.

A direct calculation shows that **all three boundaries meet at a single point** $(\alpha^*, C/R^*)$ where $\alpha^* = (p - \delta e)/[\delta(1-e)]$ and $C/R^* = e - p$. (This is forced by transitivity: if smart = rule [B1] and human = rule [B2] both hold at a point, then smart = human [B3] automatically holds there too.) The three regions are: rules in the upper-left (high C/R, low α), humans in the upper-right (high α), smart machines below the diagonal.

We draw the diagram at $e = 0.65$, which gives $\alpha^* = (0.6 - 0.4875)/[0.75 \cdot 0.35] \approx 0.43$ and $C/R^* = 0.05$.



Reading the figure. Solid segments are *binding* (they actually separate two regions); dashed segments are *not binding* (they extend the line into territory where some other boundary already settles the comparison).

- In the upper-left (high coding cost, low mitigation), **rules dominate**: coding the smart machine is too expensive, and the human's mitigation is too weak to make the delay worthwhile.
- In the upper-right (high mitigation, high coding cost), **humans dominate**: their mitigation pays off, and the alternative (smart machine) is either too costly to code or beaten on accuracy-with-mitigation.
- Below the diagonal (low coding cost), **smart machines dominate**: their speed advantage and cheap coding outweigh whatever the human can offer.

What happens as e rises? Differentiating each boundary:

- The horizontal $C/R = e - p$ shifts *up* (the rule ceiling rises — smart-vs-rule region expands).
- The vertical $\alpha^* = (p - \delta e)/[\delta(1 - e)]$ shifts *left* (the rule-vs-human boundary moves left — humans take over more former-rule territory at the top).
- The diagonal $C/R = e(1 - \delta) - \alpha\delta(1 - e)$ shifts *up and flattens* (its vertical intercept rises and slope $-\delta(1-e)$ approaches zero — smart-vs-human region expands into human territory at the bottom).

The combination: rules lose territory on both sides; smart machines expand into both former-rule and former-human territory; the human region shifts, gaining some former-rule territory at the top while losing some former-human territory at the bottom. **Whether humans gain or lose *net* territory in $(\alpha, C/R)$ space as e rises depends on parameters and is not pinned down by the model.** (Numerically at our parameters, the human region first gains area as e rises from 0.65 and then loses area for $e \gtrsim 0.73$ — the vertical-line shift dominates initially, the diagonal shift dominates later.)

What *is* unambiguous, and what Gans calls the §6.3 headline result, is a **point-by-point claim**: fix any $(\alpha, C/R)$ in the human region close to the smart-vs-human diagonal. As e rises, the

diagonal shifts up past that point and the assignment flips from human to smart machine. **That** is "displacement at the margin" — displacement at a specific fixed task as the AI improves on that task, not a claim about the global size of the human region.

The model is silent on the global-size question because Figure 6.1's "area" depends on an arbitrary C/R box and on how economic mass is distributed across $(\alpha, C/R)$ — neither of which the model specifies. To get a clean global-size claim, we need a model where the regions are indexed by something the model *does* specify. That is what Part II provides, by parameterizing tasks along a single complexity dimension.

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8 A Global Question the Model Doesn't Answer Without Help

Part I's §6.3 model is point-by-point: at any single $(\alpha, C/R, e)$, it tells you who decides. The headline result — that as e rises, smart machines push humans out at the smart-vs-human boundary — is a *point-by-point* claim about specific tasks near that boundary.

It is tempting to read this as a global claim: "as e rises, the human region shrinks." But the model alone doesn't license that reading, because the size of a region depends on a measure over $(\alpha, C/R)$, and the model doesn't supply one. (We saw earlier that even at uniform measure on $(\alpha, C/R)$ with our parameters, the human area first *grows* as e rises and only later shrinks — exactly opposite the naive reading.) The radiologist puzzle (Hinton 2016: "stop training radiologists"; 2026: more radiologists than ever) is the cost of mistaking the point-by-point claim for a global one.

Part II makes one specific commitment that lets us compute a clean global answer. We index decisions by complexity $j \in [0, 1]$, let AI accuracy fall linearly in j, and compute the human region's *width* in j as the AI improves. Both the geometry and the result are inherited from Part I — this isn't a new model. The point of doing it is to make the global behavior explicit and tractable.

\begin{center} {**Part II: Computing the Global Picture**} \end{center>

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9 The Variant

Index decision types by complexity $j \in [0, 1]$. Higher j means a harder decision. The §6.3 primitives become functions of j:

- **AI accuracy** $e(j)$, decreasing in j. AI is more accurate on simpler decisions.
- **Match payoff** R and **mismatch payoff** 0 are still constants.
- **Coding cost** C, **delay discount** δ, **prior** p, and **human mitigation** α remain constants that do not depend on j.

For tractability, take the linear specification

$$e(j) = 1 - kj, \quad k \in (0, 1],$$

so that $e(0) = 1$ (the AI is perfect on the easiest tasks) and $e(j)$ falls linearly with complexity. The parameter k is the **AI's complexity-decay rate**: how fast accuracy falls as decisions get harder. **Smaller k means a better AI everywhere.** Comparative statics in k model what happens when "the AI improves."

The three regime payoffs at decision type j:

Regime	Payoff at decision type j
Rule / dumb machine	pR
Smart machine	$(1 - kj)R - C$
Human (with AI)	$\delta R[1 - kj(1 - \alpha)]$

Three things to notice:

- The smart-machine payoff is *decreasing* in j: AI accuracy falls with complexity.
- The human payoff is also decreasing in j, but less steeply: the human catches a fraction α of the additional AI errors.
- The rule payoff is *flat* in j: the rule doesn't know or care about complexity.

10 Three Cutoffs in j

The §6.3 boundaries from Part I now specialize to cutoffs in j. We solve each pairwise indifference for j.

Smart machine vs. rule. Smart beats rule at j iff $(1-kj)R - C \geq pR$, i.e.,

$$\boxed{j \leq j_{SR}(k) \equiv \frac{1-p-C/R}{k}.}$$

Human vs. rule. Human beats rule at j iff $\delta R\big[1-kj(1-\alpha)\big] \geq pR$, i.e.,

$$\boxed{j \leq j_{HR}(k) \equiv \frac{\delta - p}{\delta\, k\,(1-\alpha)}.}$$

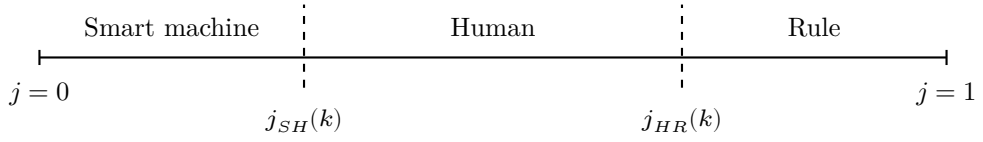
(Meaningful when $\delta > p$, which we assume throughout.)

Smart machine vs. human. Smart beats human at j iff $(1-kj)R - C \geq \delta R\big[1-kj(1-\alpha)\big]$. Rearranging:

$$\boxed{j \leq j_{SH}(k) \equiv \frac{1-\delta-C/R}{k\,[1-\delta(1-\alpha)]}.}$$

In the parameter region of interest — where $j_{SH}(k) < j_{HR}(k)$, which holds at our running parameters — the assignment ladder is:

Assignment ladder (at fixed k)



As decisions get harder, we move along the ladder: smart machine, then human, then rule.

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11 A Concrete Calculation

At our running parameters ($p = 0.6$, $\delta = 0.75$, $\alpha = 0.7$, $C/R = 0.10$), the cutoff coefficients simplify:

$$j_{SR}(k) = \frac{0.30}{k}, \qquad j_{HR}(k) = \frac{0.15}{0.225\,k} \approx \frac{0.667}{k}, \qquad j_{SH}(k) = \frac{0.15}{0.775\,k} \approx \frac{0.194}{k}.$$

For example, at $k = 1.5$ (a moderately good AI: accuracy falls from 1 at $j = 0$ to -0.5 at $j = 1$, so meaningfully predictive on the easier half of tasks):

$$j_{SH}(1.5) \approx 0.129, \qquad j_{HR}(1.5) \approx 0.444.$$

So at $k = 1.5$:

- For $j \in [0, 0.13]$: smart machine handles the decision.
- For $j \in [0.13, 0.44]$: human handles the decision.
- For $j \in [0.44, 1]$: the decision falls to the rule (or is not attempted at all).

The human handles a region of width ≈ 0.31 of the unit interval at this AI quality.

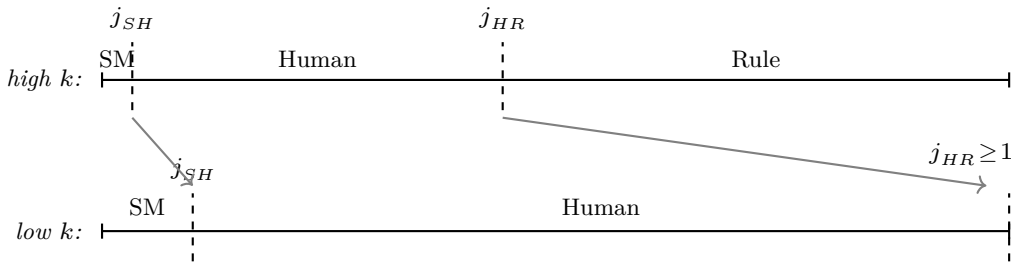
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12 What Happens as the AI Improves

Now suppose the AI improves uniformly: k falls. From the boxed expressions, **all three cutoffs scale as** $1/k$:

$$j_{SR}(k) \propto \frac{1}{k}, \quad j_{HR}(k) \propto \frac{1}{k}, \quad j_{SH}(k) \propto \frac{1}{k}.$$

So all three cutoffs shift right in unison as k decreases.



All three cutoffs scale as $1/k$, so all three shift right in unison as k decreases. The smart-machine region expands from below, and the human-rule boundary shifts right, lifting humans into former-rule territory at the top. Both effects are Part I's boundary motions, integrated along j.

The interesting question is what happens to the *width* of the human region, $j_{HR}(k) - j_{SH}(k)$. While $j_{HR}(k) < 1$, this width also scales as $1/k$ and grows as the AI improves. Once $j_{HR}(k) \geq 1$ — once the AI is good enough that humans handle every task complexity up to the top of the spectrum — only the lower cutoff continues to move, and the human region begins to shrink.

This is the **biphasic pattern**: an *expansion phase* in which the human region grows as the AI improves, followed by a *contraction phase* once the upper bound binds. The empirical content is that **the radiologist puzzle is naturally a phase-1 phenomenon**: humans end up handling more (not fewer) tasks as the AI gets better, until the AI reaches the top of the complexity spectrum — at which point further improvement crowds humans out. Whether a given industry is in phase 1 or phase 2 is an empirical question; the model doesn't pick.

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13 Numerical Example: The Biphasic Pattern

At our running parameters, here are the cutoffs at seven values of k:

k	$j_{SH}(k)$	$j_{HR}(k)$	Width of human region
3.0	0.065	0.222	0.158
2.0	0.097	0.333	0.237

k	$j_{SH}(k)$	$j_{HR}(k)$	Width of human region
1.5	0.129	0.444	0.315
1.0	0.194	0.667	0.473
0.7	0.276	0.952	0.676
0.5	0.387	1.000*	0.613
0.3	0.645	1.000*	0.355

* Capped at the upper bound of the task spectrum.

The pattern is **biphasic**:

- **Early in AI improvement** (large k, declining): the human region *expands*. AI is poor enough across the spectrum that improvement opens new task territory at the top faster than smart machines eat the bottom. From $k = 3$ to $k = 0.7$, the human region grows from 0.16 to 0.68 — more than a fourfold expansion.
- **Late in AI improvement** (small k): once j_{HR} hits 1, there are no more tasks to push into. Further improvement only erodes the human region from below. From $k = 0.7$ to $k = 0.3$, the human region shrinks from 0.68 to 0.36.

The peak of the human region is at $k^* \approx 0.67$ (where j_{HR} first hits 1). For k above the peak, the human region is in *expansion phase*; for k below the peak, it is in *contraction phase*.

Whether a given industry is in phase 1 or phase 2 depends on how complex its decisions can get and whether j_{HR} has hit its cap. Medicine, law, and engineering — with effectively unbounded task complexity — can stay in phase 1 for a long time. Domains with naturally bounded complexity (image labeling, transcription) move into phase 2 quickly.

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14 Proposition

Proposition. In the linear specification $e(j) = 1 - kj$ with constants p, δ, α, C, R satisfying $\delta > p$ and $1 - \delta - C/R > 0$, the assignment cutoffs $j_{SH}(k)$ and $j_{HR}(k)$ each scale as $1/k$. Therefore, as the AI improves (k decreases):

1. The smart-machine region $[0, j_{SH}(k)]$ expands.
2. The human region $[j_{SH}(k), j_{HR}(k)]$ shifts upward in complexity, and grows in width as long as $j_{HR}(k) < 1$.
3. Once $j_{HR}(k) \geq 1$, the human region begins to shrink (the rule region having vanished from above).

In particular, the human region under uniform j-measure follows a biphasic pattern as the AI improves: expansion first, then contraction once task-complexity saturation binds. Both phases are present in Part I's geometry as well; Part II makes them measurable.

Proof.

Part 1. $j_{SH}(k) = \frac{1-\delta-C/R}{k\,[1-\delta(1-\alpha)]}$. Since $1 - \delta - C/R > 0$ by assumption and $1 - \delta(1-\alpha) > 0$ since $\delta < 1$, $j_{SH}(k)$ is strictly decreasing in k, hence strictly increasing as k decreases. □

Part 2. $j_{HR}(k) = \frac{\delta - p}{\delta k(1-\alpha)} > 0$ since $\delta > p$ and $\alpha < 1$. Differentiating with respect to k: $dj_{HR}/dk < 0$. As k decreases, j_{HR} rises. The width of the human region is

$$j_{HR}(k) - j_{SH}(k) = \frac{1}{k}\left[\frac{\delta - p}{\delta(1-\alpha)} - \frac{1 - \delta - C/R}{1 - \delta(1-\alpha)}\right].$$

The bracketed term is a positive constant (in the parameter region of interest), so the width also scales as $1/k$ and grows as k decreases — until $j_{HR}(k)$ hits 1. □

Part 3. Once $j_{HR}(k) \geq 1$, the human region is bounded above by $j = 1$, so it equals $[j_{SH}(k), 1]$ with width $1 - j_{SH}(k)$. As k continues to fall, $j_{SH}(k)$

continues to rise, so the width shrinks. □

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15 Caveat: This Is Not a Complementarity Story

The previous handout (*Judgment and Prediction*) proved that prediction and judgment are **complements** in the Chapter 5 sense: the cross-partial of payoffs is positive — better prediction raises the marginal return to having more judgment, on a *fixed* decision. The biphasic pattern here is *not* such a story. At any fixed j, the §6.3 logic still applies: AI substitutes for the human, and the cross-partial of the human payoff in e and α is negative. The expansion phase happens because the same $e \uparrow$ that displaces humans on simple decisions makes human-with-AI viable on harder decisions where it previously wasn't — two substitution effects, integrated. We will return to genuine within-task complementarity stories later in the course.